



DEPARTMENT OF CSE-CYBER SECURITY ENGINEERING

MID SEMESTER EXAMINATION –II (April- 2026)

Subject – Mobile and Wireless Security

Subject Code – CY 803(A)

Duration: 2:00 Hours

Maximum Marks: 42

Enrol. No.....

Note: Attempt all parts.

PART A

Question			Marks	COs
Objective Type Questions (attempt all questions)				
1.	GPRS stands for:		0.5	3
	a. General Packet Radio Service	b. General Packet Radio Services		
	c. General Packet Radio Systems	d. Global Packet Radio Service		
2.	Which technology is used in 3G networks for mobile communication?		0.5	3
	a. GSM	b. LTE		
	c. UMTS	d. Bluetooth		
3.	4G networks mainly provide:		0.5	3
	a. Voice only	b. Low speed data		
	c. High speed internet and multimedia services	d. Only SMS services		
4.	RFID is mainly used for:		0.5	3
	a. Image processing	b. Video compression		
	c. Object identification and tracking	d. Data encryption		
5.	Which component is used to read RFID tags?		0.5	3
	a. Transmitter	b. Modem		
	c. RFID Reader	d. Router		
6.	Which network provides higher speed than 3G?		0.5	3
	a. 2G	b. 4G		
	c. 3G	d. GSM		
7.	LTE stands for:		0.5	4
	a. Long Term Evolution	b. Long Time Encryption		
	c. Local Transfer Evolution	d. Long Term Extension		
8.	Which type of security encrypts data between two nodes of a network?		0.5	4
	a. End-to-end security	b. Link encryption		
	c. Firewall security	d. Packet filtering		
9.	Which attack attempts to make a service unavailable?		0.5	4
	a. Phishing	b. Denial of Service		
	c. Sniffing	d. Spoofing		
10.	Authentication ensures:		0.5	4
	a. Data compression	b. Identity verification		
	c. Data storage	d. Data formatting		
11.	Nonrepudiation means:		0.5	4
	a. Preventing data theft	b. Sender cannot deny sending the message		
	c. Encryption of data	d. Network isolation		
12.	Which protocol version provides improved internet security features?		0.5	4
	a. IPv3	b. IPv4		
	c. IPv5	d. IPv6		

	E-commerce refers to:		0.5	5
	a. Electronic manufacturing	b. Online gaming		
	c. Buying and selling goods through the internet	d. Computer networking		
14	M-commerce stands for:		0.5	5
·	a. Mobile commerce	b. Multi commerce		
	c. Modern commerce	d. Machine commerce		
15	Which of the following is an electronic payment method?		0.5	5
·	a. Credit card	b. Debit card		
	c. Digital wallet	d. All of the above		
16	Secure Electronic Transaction (SET) mainly provides:		0.5	5
·	a. Network routing	b. Secure online payment transactions		
	c. Video streaming	d. Data compression		
17	Encryption is used to:		0.5	5
·	a. Increase internet speed	b. Protect data confidentiality		
	c. Reduce file size	d. Delete files		
18	Integrity in electronic transactions ensures:		0.5	5
·	a. Data remains unchanged	b. Data is compressed		
	c. Data is deleted	d. Data is duplicated		

PART B

Short Answer Type Questions (Attempt any two)

1.	What is GPRS? Explain its role in mobile communication.	2	3
2.	Write the key features of 4G networks.	2	3
3.	What is RFID technology? Mention its applications.	2	3

Short Answer Type Questions (Attempt any two)

4.	What is LTE technology? Mention its advantages.	2	4
5.	Define authentication and privacy in wireless networks.	2	4
6.	What is a Denial of Service (DoS) attack?	2	4

Short Answer Type Questions (Attempt any two)

7.	Define E-commerce and M-commerce.	2	5
8.	What are electronic payment systems?	2	5
9.	Write the importance of encryption in secure transactions.	2	5

PART C

Q. No.	Question	Bloom's Taxonomy Level	Marks	COs
1.	a) Explain the architecture and working of UMTS and its advantages in mobile communication. OR b) Compare 3G and 4G networks in terms of speed, technology, and applications.	Understand (2) Analyse (4)	7	3
2.	a) Explain different security models in wireless networks including military and civil security models. OR b) Discuss end-to-end security and link encryption with suitable examples.	Understand (2) Analyse (4)	7	4
3.	a) Explain different types of electronic payment systems and electronic cards used in e-commerce. OR b) Discuss Secure Electronic Transaction (SET) and its security requirements such as trust, encryption, authentication, confidentiality, integrity, and nonrepudiation.	Understand (2) Analyse (4)	7	5

About CO column: It stands for Course Outcome.

Course Outcome Statements:

CO3	Explain the working and applications of 3G/4G networks, GPRS, UMTS and RFID.
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CO4	Analyze security issues and protection mechanisms in wireless networks.
CO5	Evaluate e-commerce and m-commerce systems including electronic payments and secure transactions.